

Artificial Intelligence & Machine Learning Course Outline

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Prepared For: Student & Parents

Course Overview

This course is designed to provide students with a strong foundation in Artificial Intelligence, Data Science, and Machine Learning. The training program focuses on both theoretical understanding and practical implementation using industry-standard tools and real-world projects.

The curriculum is divided into multiple phases, beginning with foundational concepts and progressing toward advanced Machine Learning and Deep Learning topics.

1. Phase 1: Building the Foundations

1.1. Introduction to Artificial Intelligence

- Introduction to Artificial Intelligence (AI)
- Real-world applications of AI
- Overview of:
 - Machine Learning (ML)
 - Deep Learning (DL)
 - Artificial Neural Networks (ANN)
 - Convolutional Neural Networks (CNN)

1.2. Data Analysis and Visualization

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Data cleaning and preprocessing
- Data visualization techniques

1.3. Database & Parallel Computing Concepts

- Introduction to SQLite3
- Multithreading
- Multiprocessing

- Importance of parallel processing in handling large datasets and model training

1.4. Statistics Fundamentals

- Introduction to Statistics
- Types of Statistics
- Population and Sample Data
- Measures of Central Tendency
- Measures of Dispersion
- Variables and Data Types
- Correlation and Covariance

1.5. Probability

- Addition Rules of Probability
 - Mutually Exclusive Events
 - Non-Mutually Exclusive Events
- Multiplication Rules of Probability
 - Independent Events
 - Dependent Events

1.6. Probability Distributions

- PDF, PMF, and CDF
- Binomial Distribution
- Poisson Distribution
- Normal Distribution
- Z-Score

1.7. Inferential Statistics

- Hypothesis Testing
- P-Value
- Z-Test
- T-Test
- Type I and Type II Errors
- Bayes Theorem
- Chi-Square Test

1.8. Feature Engineering

- Handling Missing Values
- Handling Imbalanced Datasets
- Outlier Detection and Treatment
- Encoding Techniques
 - Nominal Encoding
 - One-Hot Encoding
 - Ordinal Encoding

1.9. Exploratory Data Analysis (EDA)

- Understanding datasets
- Data cleaning and transformation
- Visual exploration of data
- Practice on publicly available datasets

2. Phase 2: Machine Learning

2.1. Introduction to Machine Learning

- Types of Machine Learning
- Machine Learning Pipeline
- Model training and evaluation workflow

2.2. Regression Techniques

- Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Cost Functions
- Performance Metrics
 - MSE
 - MAE
 - RMSE
- Overfitting and Underfitting

2.3. Regularization Techniques

- Ridge Regression
- Lasso Regression
- ElasticNet Regression

2.4. Classification Algorithms

- Logistic Regression
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

2.5. Tree-Based Algorithms

- Decision Tree Classifiers and Regressors
- Random Forest
- AdaBoost
- Gradient Boosting
- XGBoost

2.6. Project-Based Learning

- Hands-on projects with each major algorithm
- Real-world dataset implementation
- End-to-end Machine Learning projects

3. Future Phases (Advanced Topics)

The upcoming advanced phases of the course will include:

- Clustering Algorithms
- MLOps
- Natural Language Processing (NLP)
- Deep Learning
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory Networks (LSTM)
- Transformers and Modern AI Architectures

Detailed outlines for advanced phases will be shared separately.

Teaching Methodology

- Conceptual learning with practical implementation
- Hands-on coding sessions
- Real-world datasets and projects
- Regular assignments and practice exercises
- Industry-oriented learning approach

Course Objective

The primary objective of this course is to equip students with strong analytical, programming, and problem-solving skills required for modern AI and Machine Learning applications.

Prepared as a professional learning roadmap for students and parents.